

Column Bounds for the Circuit Imbalance Measure

Neta Singer

École Polytechnique Fédérale de Lausanne

Joint work with D. Dadush, F. Eisenbrand, R. Pinchasi, and T. Rothvoss.

September 5, 2024

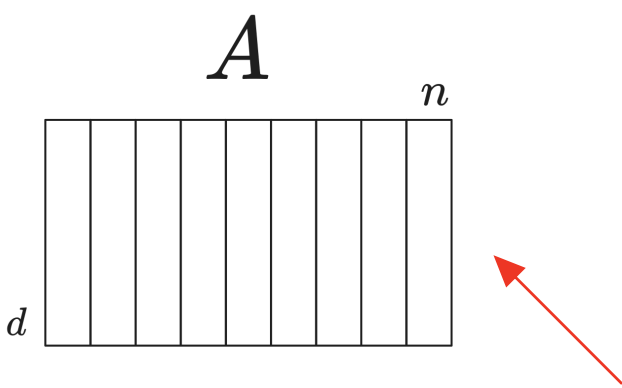
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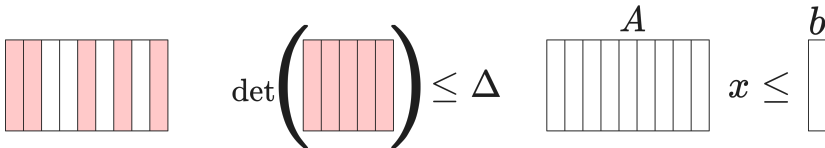
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Complexity Measures for Matrices

- **Δ -modular matrices:** $A \in \mathbb{Z}^{d \times n}$, $|\det(B)| \leq \Delta$ for any B a $d \times d$ submatrix of A .

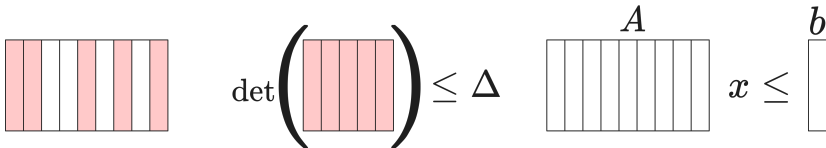


$$\det \left(\begin{array}{cccc} \text{red} & \text{red} & \text{red} & \text{red} \end{array} \right) \leq \Delta$$

$$A \quad x \leq b$$

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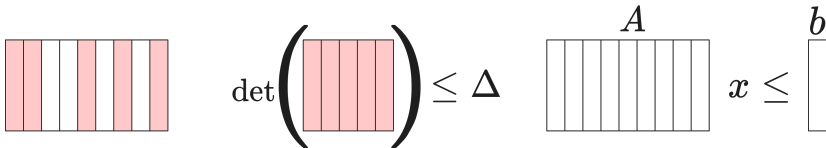
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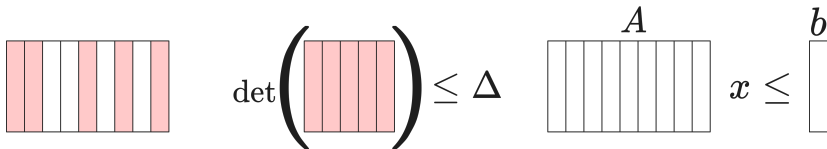
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- Δ -modular integer programs with row and column support at most 2 can be solved in poly time [FJWY'2021]

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- 2-modular integer programs can be solved in poly time [AWZ'2017]
- Δ -modular integer programs with row and column support at most 2 can be solved in poly time [FJWY'2021]
- Can integer programs be solved in poly time for any constant Δ ?

Complexity Measures for Matrices

How many distinct (or non-parallel) columns can such a Δ bounded matrix have?

Matrix type	Large d bounds		Large Δ bounds
TU Matrix	$\frac{1}{2}(d^2 + d)$		
Δ -modular	$\Delta^{\log_2 \log_2 \Delta + 2} d^2$	[GWZ'18]	$O(d^4 \Delta)$ [AS'23]
	$d^2 + f(\Delta)d$	[GNW'21]	$\frac{(d^2 + d)\Delta^2}{2}$ [LPSX'23]
Δ -modular (non-parallel)	$\binom{d}{2} + 80\Delta^7 d$	[PSWX'24]	

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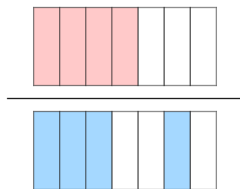
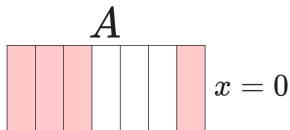
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- **Circuit imbalance measure:**

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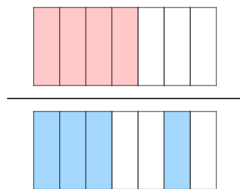
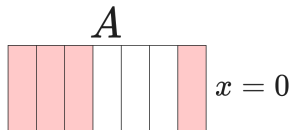


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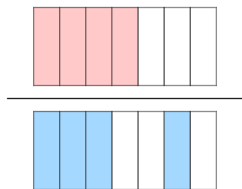
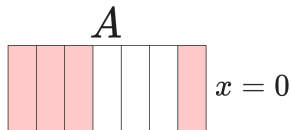
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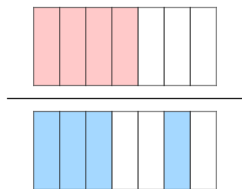
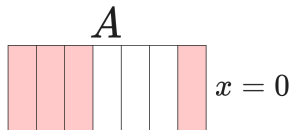
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- **Question:** How many non-parallel columns can such a κ -bounded matrix have?

Our Results

Theorem (DEPRS 2024+)

For $A \in \mathbb{R}^{d \times n}$ a matrix with non-collinear columns, $n \leq \pi d^4 \kappa_A$.

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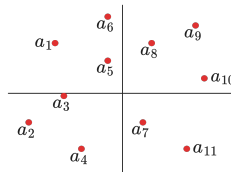
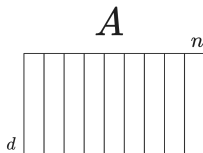
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Lemma (DEPRS 2024+)

For $l \geq 2$, a simple rank d complex representable matroid excluding $U_{2,l}$ as a minor has at most $\frac{1}{4}d^4 l$ elements.

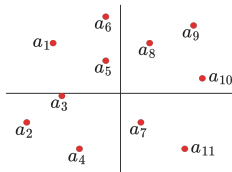
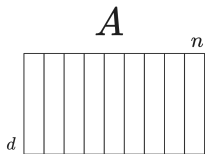
Reduction to hyperplane arrangements

- $A \in \mathbb{R}^{d \times n}$, $\text{rank}(A) = d$.

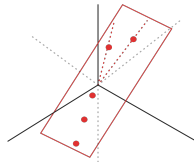
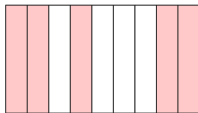


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- A_B a basis of A , $\text{span}(A_{B \setminus \{i\}})$ hyperplane.



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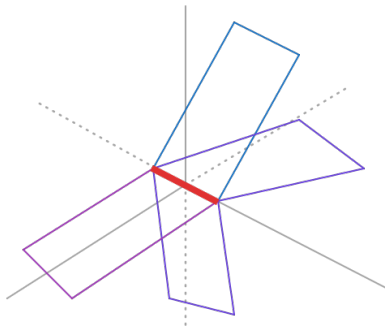
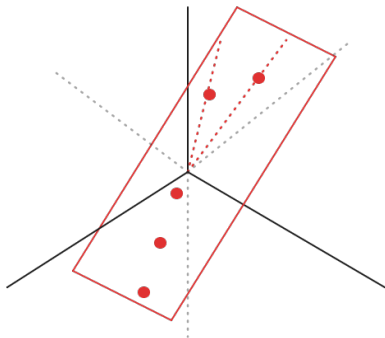
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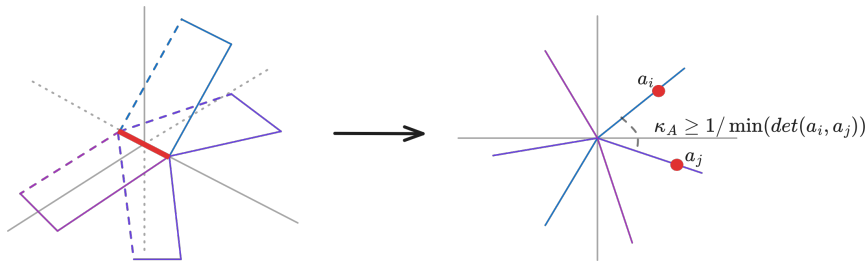
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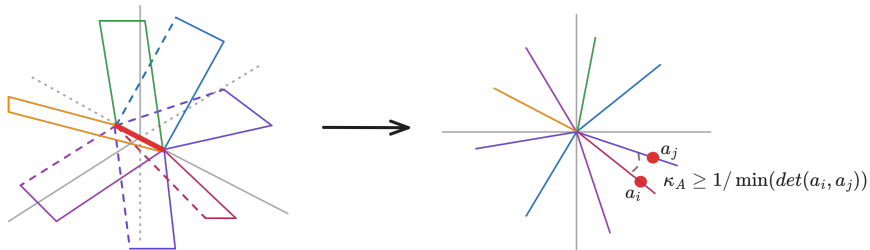
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Reduction to MaxLines estimates

- **Want to find:** $H_1, \dots, H_l$ spanned by A s.t. $H_i \supset F$ for F a $(d - 2)$ -dim (affine) subspace.

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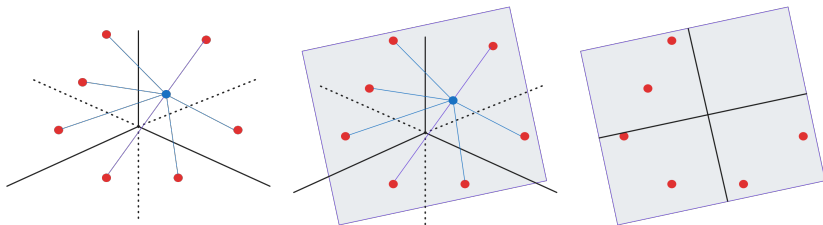
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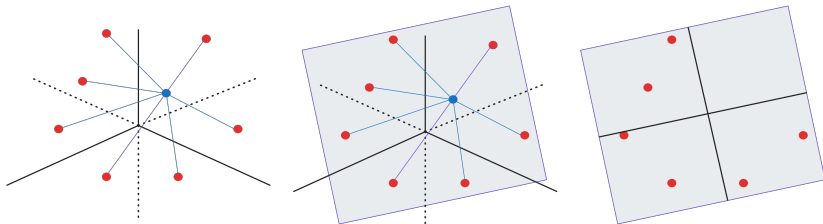
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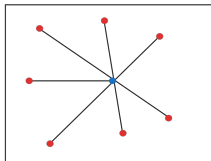
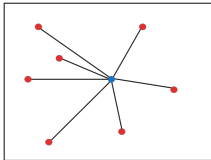
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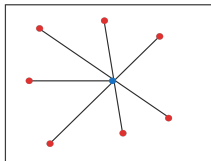
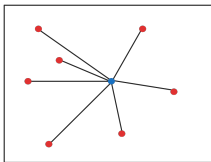
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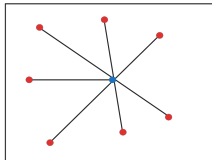
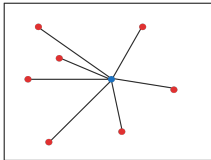
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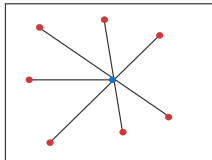
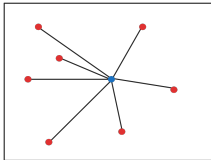
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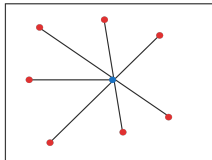
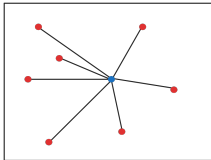
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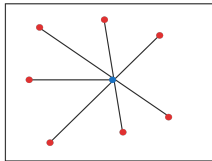
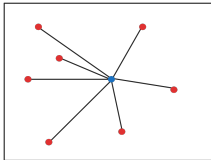
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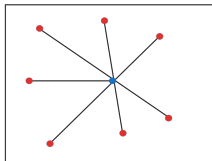
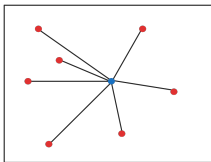
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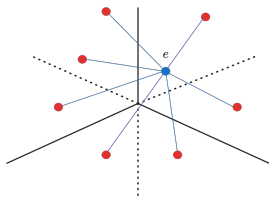
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 - **Our improvement:** for full-dimensional $S \subset \mathbb{R}^d$, $\text{maxlines}(S) \geq (1 - \frac{4}{d+1})|S|$.
- By applying the induction, the number of hyperplanes is $\geq \frac{4}{d^4}|S|$.

In Matroid Terminology

- Let $\mathcal{M} = (S, \mathcal{I})$ be a (complex)-representable affine matroid.
- A *circuit* of size 3 in $\mathcal{M}$ is a set of 3 collinear points.
- A *central projection* with center $e \in S$ is a contraction and simplification $si(\mathcal{M}/e)$.
- The *size* of $si(\mathcal{M}/e)$ is at least $\text{maxlines}(S)$.
- A rank d matroid has $\text{maxlines}(S) \geq (1 - \frac{4}{d+1})|S|$.

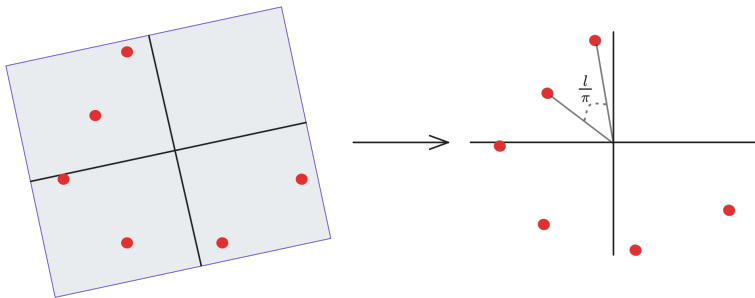


Lemma (DEPRS 2024+)

A simple rank d complex representable matroid excluding $U_{2,l}$ as a minor has at most $\frac{1}{4}d^4l$ elements.

2-D base case for κ

- A rank d matroid $\mathcal{M} = (S, \mathcal{I})$ has a minor of rank 2 of size $\frac{4}{d^4}|S|$.
- Contraction and simplification only decreases κ .
- Lower bounds on κ for a rank 2 matroid of size l carry over to lower bounds over $\mathcal{M}$.
- Base case given by $\kappa \geq \frac{l}{4\pi}$.
- Finally, $\pi d^4 \kappa \geq |S|$.



Open Questions

1. Can the constant $(1 - \frac{4}{d+1})$ for the $\text{maxlines}(S)$ estimate be improved?
2. Are there other excluded minors in terms of κ ?
3. What happens for other types of matroids?

Thank you!

References

-  Stephan Artmann, Robert Weismantel, and Rico Zenklusen (2017)
A strongly polynomial algorithm for bimodular integer linear programming.
STOC pages 1206–1219.
-  Samuel Fiorini, Gwenaël Joret, Stefan Weltge, and Yelena Yuditsky (2021)
Integer programs with bounded subdeterminants and two nonzeros per row.
FOCS pages 13–24.
-  Christoph Glanzer, Robert Weismantel, and Rico Zenklusen (2018)
On the number of distinct rows of a matrix with bounded subdeterminants.
SIAM Journal on Discrete Mathematics pages 1706–1720.
-  Jim Geelen, Peter Nelson, and Zach Walsh (2021)
Excluding a line from C-representable matroids.
arXiv preprint arXiv:2101.12000.



Gennadiy Averkov and Matthias Schymura (2023)

On the maximal number of columns of a δ -modular integer matrix: bounds and computations.
Mathematical Programming pages 1-29.



Jon Lee, Joseph Paat, Ingo Stallknecht, and Luze Xu (2023)

Polynomial upper bounds on the number of differing columns of δ -modular integer programs.
Mathematics of Operations Research 48(4):2267–2286.



Joseph Paat, Ingo Stallknecht, Zach Walsh, and Luze Xu (2024)

On the column number and forbidden submatrices for-modular matrices.
SIAM Journal on Discrete Mathematics 38(1):1–18.



Zeye Han (2017)

A note on weak dirac conjecture.



Zeev Dvir, Shubhangi Saraf, and Avi Wigderson (2014)

Improved rank bounds for design matrices and a new proof of kelly's theorem.
Forum of Mathematics, Sigma volume 2, page e4.